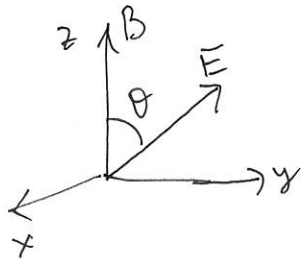
Relativistic motion of charged particles in constant EM field.

$$m\frac{du^\alpha}{d\tau} = q\, F^{\alpha\beta} u_\beta$$

$$\frac{d}{d\tau}\begin{pmatrix} u_0 \\ u_x \\ u_y \\ u_z \end{pmatrix} = \frac{q}{m}\begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}\begin{pmatrix} u_0 \\ -u_x \\ -u_y \\ -u_z \end{pmatrix}$$



Denote plane made by $\vec{B}$ & $\vec{E}$ (yz).

$$\frac{du^\alpha}{d\tau} = \frac{q}{m} F^{\alpha\beta} u_\beta = \frac{q}{m} F^{\alpha\gamma} g_{\gamma\beta} u^\beta$$

$$F^{\alpha\beta} = \begin{pmatrix} 0 & 0 & -E\sin\theta/c & -E\cos\theta/c \\ 0 & 0 & -B & 0 \\ E\sin\theta/c; & B & 0 & 0 \\ E\cos\theta/c; & 0 & 0 & 0 \end{pmatrix}$$

$$\frac{du^0}{d\tau} = \frac{q}{m} F^{0\beta} u_\beta = \frac{q}{m}\frac{E}{c}(\sin\theta\, u_y + \cos\theta\, u_z)$$

$$\frac{du_x}{d\tau} = \frac{q}{m} F^{1\beta} u_\beta = \frac{qB}{m} u_y$$

$$\frac{du_y}{d\tau} = \frac{q}{m} F^{2\beta} u_\beta = -\frac{qE}{mc}\sin\theta\, u_0 - \frac{qB}{m} u_x$$

$$\frac{du_z}{d\tau} = \frac{q}{m} F^{3\beta} u_\beta = \frac{q}{m}\frac{E}{c}\cos\theta\, u_0 \qquad \Lambda = \frac{qE}{mc},\ \omega = \frac{qB}{m}$$

$$\Rightarrow \frac{d}{d\tau}\begin{pmatrix} u_0 \\ u_x \\ u_y \\ u_z \end{pmatrix} = \begin{pmatrix} 0 & 0 & \Lambda\sin\theta & \Lambda\cos\theta \\ 0 & 0 & \omega & 0 \\ \Lambda\sin\theta & -\omega & 0 & 0 \\ \Lambda\cos\theta & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} u_0 \\ u_x \\ u_y \\ u_z \end{pmatrix} \Rightarrow \frac{d\vec{u}}{d\tau} = M\vec{u}$$

this matrix is $\frac{q}{m} F^{\alpha\gamma} g_{\gamma\beta}$

$$u \sim e^{\lambda \tau}$$

$$\lambda \vec{u} = M \vec{u} \quad \Rightarrow \quad \det(M - \lambda) = 0$$

$$\det \begin{bmatrix} -\lambda & 0 & \Lambda \sin\theta & \Lambda \cos\theta \\ 0 & -\lambda & \omega & 0 \\ \Lambda \sin\theta & -\omega & -\lambda & 0 \\ \Lambda \cos\theta & 0 & 0 & -\lambda \end{bmatrix} = 0$$

$$\cancel{+\lambda \begin{bmatrix} -\lambda & 0 & \Lambda \sin\theta \\ 0 & -\lambda & \omega \end{bmatrix}}$$

$$\cancel{-\lambda \begin{bmatrix} 0 & \Lambda \sin\theta & \Lambda \cos\theta \\ -\lambda & \omega & 0 \end{bmatrix}}$$

$$-\lambda \begin{pmatrix} -\lambda & \omega & 0 \\ -\omega & -\lambda & 0 \\ 0 & 0 & -\lambda \end{pmatrix} + \Lambda \sin\theta \begin{pmatrix} 0 & -\lambda & 0 \\ \Lambda \sin\theta & -\omega & 0 \\ \Lambda \cos\theta & 0 & -\lambda \end{pmatrix} - \Lambda \cos\theta \begin{pmatrix} 0 & -\lambda & \omega \\ \Lambda \sin\theta & -\omega & -\lambda \\ \Lambda \cos\theta & 0 & 0 \end{pmatrix} =$$

$$= (-\lambda) \left\{ (-\lambda) \begin{vmatrix} -\lambda & 0 \\ 0 & -\lambda \end{vmatrix} - \omega \begin{vmatrix} -\omega & 0 \\ 0 & -\lambda \end{vmatrix} \right\}$$

$$+ \Lambda \sin\theta \left\{ \lambda \begin{vmatrix} \Lambda \sin\theta & 0 \\ \Lambda \cos\theta & -\lambda \end{vmatrix} \right\} - \Lambda \cos\theta \left\{ \lambda \begin{vmatrix} \Lambda \sin\theta & -\lambda \\ \Lambda \cos\theta & 0 \end{vmatrix} + \omega \begin{vmatrix} \Lambda \sin\theta & -\omega \\ \Lambda \cos\theta & 0 \end{vmatrix} \right.$$

$$= \lambda^4 + \lambda^2 \omega^2 - (\Lambda \sin\theta)^2 \lambda^2 - (\Lambda \cos\theta)^2 \lambda^2 - (\Lambda \cos\theta)^2 \omega^2$$

$$= \lambda^4 + \lambda^2 (\omega^2 - \Lambda^2) - \Lambda^2 \omega^2 \cos^2\theta = 0$$

$$\lambda^2 = \frac{1}{2} \left[\Lambda^2 - \omega^2 \pm \sqrt{(\Lambda^2 - \omega^2)^2 + 4 \Lambda^2 \omega^2 \cos^2\theta} \right]$$

$$\Lambda^2 - \omega^2 = \frac{q^2}{m^2} \left(\left(\frac{E}{c}\right)^2 - B^2 \right) = -\frac{q^2}{2m^2} F_{\alpha\beta} F^{\alpha\beta} = \underbrace{-\frac{q^2}{2m^2}}_{= -\frac{q^2}{2m^2} I_1} \left(B^2 - E^2/c^2\right) \cdot 2$$

$$\Lambda^2 \omega^2 \cos^2\theta = \left(\frac{q}{m}\right)^4 \left(\frac{E \cos\theta}{c}\right)^2 B^2 = \left(\frac{q}{m}\right)^4 \frac{1}{16} \left(\frac{4 \vec{E} \cdot \vec{B}}{c}\right)^2 = \underbrace{\left(\frac{q}{2m}\right)^4}_{\cdot I_2}$$

$$\lambda^2 = \frac{1}{2}\left\{-\frac{q^2}{2m^2}\left(B^2 - E^2/c^2\right) \pm \sqrt{\left(\frac{q^2}{2m^2}\right)^2\left(B^2 - E^2/c^2\right)^2 + 4\left(\frac{q}{2m}\right)^4\left(\frac{4\vec{E}\cdot\vec{B}}{c}\right)^2}\right\}$$

$$= \frac{1}{2}\left\{-\frac{q^2}{4m^2} I_1 \pm \frac{q^2}{2m^2}\sqrt{I_1^2 + I_2^2}\right\} =$$

$$= -\frac{q^2}{4m^2}\left\{I_1 \mp \sqrt{I_1^2 + I_2^2}\right\} \qquad I_1 = \left(B^2 - \frac{E^2}{c^2}\right)\cdot 2$$

$$I_2 = \frac{4}{c}\,\vec{E}\cdot\vec{B}$$

λ depends on the invariants of EM field only: $I_{1,2}$.

$\Rightarrow$ $u^\alpha(\tau)$ depends on I_1, I_2 and proper time τ (also invariant!)

~~4 roots:~~ $\cancel{\lambda_{1,2} = \frac{q}{2m}\left[\sqrt{I_1^2 + I_2^2} - I_1\right]^{1/2}(\pm 1) = \pm\varkappa.}$ ~~(real)~~

$$\cancel{\lambda_{3,4} = \pm i\Omega, \quad \Omega = \frac{q}{2m}\left[\sqrt{I_1^2 + I_2^2} + I_1\right]^{1/2}}$$

Consider case of $I_2 = 0 \Rightarrow \vec{E}\cdot\vec{B} = 0 \Rightarrow$ orthogonal fields.

(a) let $\frac{E}{c} > B \Rightarrow I_1 < 0$, $I_1 = -|I_1| = -\left(\frac{E^2}{c^2} - B^2\right)\cdot 2$

$$\lambda^2 = -\frac{q^2}{4m}\left(I_1 \mp |I_1|\right) = -\frac{q^2}{4m^2}\left(-|I_1| \mp |I_1|\right) =$$

$$= \begin{cases} \frac{q^2}{2m^2}|I_1| & (-) \\ 0 & (+) \end{cases} = \begin{cases} \frac{q^2}{m^2}\left(\frac{E^2}{c^2} - B^2\right) \\ 0 \end{cases}$$

$$\Rightarrow \lambda = \pm\frac{q}{m}\sqrt{\frac{E^2}{c^2} - B^2} \quad \Rightarrow u \sim e^{\lambda\tau} \text{ grows exponentially}$$

(b) let $B > \frac{E}{c} \Rightarrow I_1 = 2\left(B^2 - \frac{E^2}{c^2}\right) > 0 \Rightarrow |I_1| = I_1$

$$\lambda^2 = -\frac{q^2}{4m^2}\left(I_1 \mp I_1\right) = \begin{cases} 0 & (-) \\ -\frac{q^2}{2m^2} I_1 & (+) \end{cases} = \begin{cases} 0 & (-) \\ -\frac{q^2}{m^2}\left(B^2 - \frac{E^2}{c^2}\right) & (+) \end{cases}$$

$$\Rightarrow \lambda_{1,2} = \pm i\Omega, \quad \Omega = \frac{q}{m}\sqrt{B^2 - \frac{E^2}{c^2}}$$

$u \sim e^{\lambda\tau} \Rightarrow$ oscillating solutions.

Recall that there exists frame K' in which $E'_y = 0$:

$$E'_y = \gamma(E - vB) \Rightarrow v = \frac{E}{B} < c.$$

In this frame only $B'_z = \gamma\left(B - \frac{1}{c}vE\right) = \gamma\left(B - \frac{E^2}{cB}\right)$ is present.

Frame K' moves with $v = \frac{E}{B}$ along $\hat{x}$-axis.

In K', particle performs Larmor motion around B'_z.

In the lab frame K the center of that orbit drifts with $v = \frac{E}{B}$ along $\hat{x}$-axis.

$\underline{E\neq 0, B=0}$: let $\vec{E}$ be constant, and particle initial energy $\mathcal{E}_0$, momentum $\vec{p}_0$.
Initially the particle is at the origin.

$\Rightarrow I_2 = 0 \;,\; I_1 = 2\left(B^2 - \frac{E^2}{c^2}\right) = -\frac{2}{c^2}E^2 < 0.$

Choose $\vec{E}$ to be along $\hat{z}$ axis ($\theta = 0$ so that $\cos\theta = 1, \sin\theta = 0$)

$$\Rightarrow \frac{d}{d\tau}\begin{pmatrix} u_0 \\ u_x \\ u_y \\ u_z \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & \Lambda \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \Lambda & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} u_0 \\ u_x \\ u_y \\ u_z \end{pmatrix}$$

$$\Rightarrow \quad \dot{u}_x = 0 \;,\; \dot{u}_y = 0 \;\Rightarrow\; \frac{d}{d\tau}(\gamma v_x) = 0 = \frac{d}{d\tau}(\gamma v_y)$$

Since $\vec{p} = m\gamma\vec{v} \Rightarrow \frac{d}{d\tau}(p_x) = 0 = \frac{d}{d\tau}(p_y)$

$$p_x(\tau) = p_{0x} \;,\; p_y(\tau) = p_{0y}$$

Since $u^\alpha = \frac{dx^\alpha}{d\tau}$: $\frac{du_x}{d\tau} = 0 \Rightarrow u_x(\tau) = u_{0x} = \frac{p_{0x}}{m}$

$$u_y(\tau) = u_{0y} = \frac{p_{0y}}{m}$$

$$\Rightarrow x(\tau) = u_{0x}\tau = \frac{p_{0x}}{m}\tau$$
$$y(\tau) = u_{0y}\tau = \frac{p_{0y}}{m}\tau.$$

$$\frac{d}{d\tau}\begin{pmatrix} u_0 \\ u_z \end{pmatrix} = \begin{pmatrix} 0 & \Lambda \\ \Lambda & 0 \end{pmatrix}\begin{pmatrix} u_0 \\ u_z \end{pmatrix}$$

$$u \sim e^{\lambda\tau} \Rightarrow \begin{pmatrix} -\lambda & \Lambda \\ \Lambda & -\lambda \end{pmatrix}\begin{pmatrix} u_0 \\ u_z \end{pmatrix} = 0 \Rightarrow \det\begin{pmatrix} -\lambda & \Lambda \\ \Lambda & -\lambda \end{pmatrix} = 0$$

$$\Rightarrow \lambda^2 = \Lambda^2 \Rightarrow \lambda = \pm\Lambda = \pm\frac{qE}{mc}$$

$\lambda = +\Lambda$: $\begin{pmatrix} -\Lambda & \Lambda \\ \Lambda & -\Lambda \end{pmatrix}\begin{pmatrix} u_0 \\ u_z \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} u_0 \\ u_z \end{pmatrix} = 0 \Rightarrow$

$$\Rightarrow \begin{pmatrix} u_0 \\ u_z \end{pmatrix} = a\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$\lambda = -\Lambda$:
$$\begin{pmatrix} \Lambda & \Lambda \\ \Lambda & \Lambda \end{pmatrix}\begin{pmatrix} u_0 \\ u_z \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} u_0 \\ u_z \end{pmatrix} = 0$$

$$\Rightarrow \begin{pmatrix} u_0 \\ u_z \end{pmatrix} = b\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Hence
$$\begin{pmatrix} u_0(\tau) \\ u_z(\tau) \end{pmatrix} = a\begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{\Lambda\tau} + b\begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-\Lambda\tau}$$

At $\underline{\tau = 0}$:
$$\begin{pmatrix} u_0(0) \\ u_z(0) \end{pmatrix} = \begin{pmatrix} a+b \\ a-b \end{pmatrix} = \begin{pmatrix} \frac{\mathcal{E}_0}{mc} \\ \frac{p_{0z}}{m} \end{pmatrix}$$

$$\mathcal{E}_0 = \sqrt{m^2c^4 + c^2p_0^2} \qquad u_0(0) = \gamma_0 c = \frac{\mathcal{E}_0}{mc} = \frac{1}{mc}\left(mc^2\gamma_0\right)$$

$$\vec{p}_0 = (p_{0x}, p_{0y}, p_{0z}) \qquad u_z(0) = \gamma_0 v_z(0) = \frac{p_{0z}}{m}$$

$$a = \frac{\mathcal{E}_0}{mc} + \frac{p_{0z}}{m} = \frac{\mathcal{E}_0 + p_{0z}c}{2mc}$$

$$b = \frac{\mathcal{E}_0 - p_{0z}c}{2mc}$$

$$\Rightarrow u_0(\tau) = ae^{\Lambda\tau} + be^{-\Lambda\tau} = \frac{\mathcal{E}_0 + p_{0z}c}{2mc}e^{\Lambda\tau} + \frac{\mathcal{E}_0 - p_{0z}c}{2mc}e^{-\Lambda\tau}$$

$$= \frac{\mathcal{E}_0}{mc}\operatorname{ch}\Lambda\tau + \frac{p_{0z}}{m}\operatorname{sh}\Lambda\tau$$

$$u_z(\tau) = ae^{\Lambda\tau} - be^{-\Lambda\tau} = \frac{\mathcal{E}_0}{mc}\operatorname{sh}\Lambda\tau + \frac{p_{0z}}{m}\operatorname{ch}\Lambda z.$$

But
$$u_0 = \frac{dx^0}{d\tau} = c\frac{dt}{d\tau} \Rightarrow t(\tau) = \frac{1}{c}\int u_0(\tau)\,d\tau$$

$$u_z = \frac{dx^3}{d\tau} = \frac{dz}{d\tau} \Rightarrow z(\tau) = \int u_z(\tau)\,d\tau$$

Using $\int d\tau \, ch\Lambda\tau = \frac{sh\Lambda\tau}{\Lambda}$, $\int d\tau \, sh\Lambda\tau = \frac{ch\Lambda\tau}{\Lambda}$

$$\Rightarrow \quad t(\tau) = \left\{ \frac{\mathcal{E}_0}{mc^2\Lambda} sh\Lambda\tau + \frac{p_{0z}}{mc\Lambda} ch\Lambda\tau \right\}\Bigg|_0^\tau = \qquad \Lambda = \frac{qE}{mc}$$

$$= \frac{\mathcal{E}_0}{qEc} sh(\Lambda\tau) + \frac{p_{0z}}{qE}\left(ch(\Lambda\tau) - 1\right);$$

$$z(\tau) = \left\{ \frac{\mathcal{E}_0}{mc\Lambda} ch\Lambda\tau + \frac{p_{0z}}{m\Lambda} sh\Lambda\tau \right\}\Bigg|_0^\tau =$$

$$= \frac{\mathcal{E}_0}{qE}\left(ch\Lambda\tau - 1\right) + \frac{p_{0z}c}{qE} sh(\Lambda\tau).$$

Can now express proper time τ in terms of lab time t, via explicit form of $t(\tau)$:

$$t + \frac{p_{0z}}{qE} = \frac{\mathcal{E}_0}{qEc} sh\Lambda\tau + \frac{p_{0z}}{qE} ch\Lambda\tau$$

$$p_{0z} + qEt = \frac{\mathcal{E}_0}{c} \frac{e^{\Lambda\tau} - e^{-\Lambda\tau}}{2} + p_{0z} \frac{e^{\Lambda\tau} + e^{-\Lambda\tau}}{2} =$$

$$= \left(\frac{\mathcal{E}_0}{c} + p_{0z}\right)\frac{e^{\Lambda\tau}}{2} - \left(\frac{\mathcal{E}_0}{c} - p_{0z}\right)\frac{e^{-\Lambda\tau}}{2}$$

$$\Rightarrow \quad \left(\frac{\mathcal{E}_0}{c} + p_{0z}\right) e^{2\Lambda\tau} - 2\left(p_{0z} + qEt\right) e^{\Lambda\tau} - \left(\frac{\mathcal{E}_0}{c} - p_{0z}\right) = 0$$

quatratic equ for $x = e^{\Lambda\tau}$:

$$e^{\Lambda\tau} = \frac{p_{0z} + qEt \pm \sqrt{(p_{0z} + qEt)^2 + \left(\frac{\mathcal{E}_0}{c} + p_{0z}\right)\left(\frac{\mathcal{E}_0}{c} - p_{0z}\right)^2}}{\frac{\mathcal{E}_0}{c} + p_{0z}}$$

Of course need to choose (+) sign as $e^{\Lambda\tau} > 0$ for all τ.

Thus $\tau = \frac{1}{\Lambda} \ln\left(e^{\Lambda\tau}\right)$

$$\tau = \frac{mc}{qE} \ln\left[\frac{p_{oz} + qEt + \sqrt{(p_{oz}+qEt)^2 + \frac{\mathcal{E}_o^2}{c^2} - p_{oz}^2}}{\mathcal{E}_o/c + p_{oz}}\right]$$

$$= \frac{mc}{qE} \ln\left[\frac{p_{oz} + qEt + \sqrt{(p_{oz}+qEt)^2 + m^2c^2 + p_{ox}^2 + p_{oy}^2}}{\frac{\mathcal{E}_o}{c} + p_{oz}}\right]$$

And then $z(\tau) = -\frac{\mathcal{E}_o}{qE} + \frac{\mathcal{E}_o}{2qE}\left(e^{\Lambda\tau} + e^{-\Lambda\tau}\right) + \frac{p_{oz}c}{2qE}\left(e^{\Lambda\tau} - e^{-\Lambda\tau}\right)$

$$= -\frac{\mathcal{E}_o}{qE} + \frac{\mathcal{E}_o + p_{oz}c}{2qE} e^{\Lambda\tau} + \frac{\mathcal{E}_o - p_{oz}c}{2qE} \frac{1}{e^{\Lambda\tau}}$$

$$\cancel{= -\frac{\mathcal{E}_o}{qE} + \frac{\sqrt{(p_{oz}+qEt)^2 + \frac{\mathcal{E}_o^2}{c^2} - p_{oz}^2}}{2qEc} + \frac{\mathcal{E}_o + p_{oz}c}{2qE} \frac{\left(\frac{\mathcal{E}_o}{c} + p_{oz}c\right)\frac{1}{c}}{\sqrt{(p_{oz}+qEt)^2 + \frac{\mathcal{E}_o^2}{c^2} - p_{oz}^2}}}$$

$$\cancel{= -\frac{\mathcal{E}_o}{qE} + \frac{c}{2qE}\left(p_{oz} + qEt + \sqrt{\ }\right) + \frac{c}{2qE} \frac{\frac{\mathcal{E}_o}{c} - p_{oz}}{\frac{\mathcal{E}_o}{c} + p_{oz}} \cdot}$$

$$\cancel{\cdot \frac{1}{p_{oz} + qEt + \sqrt{\ }} = -\frac{\mathcal{E}_o}{qE} + \frac{c}{2qE}\left\{\left(p_{oz} + qEt + \sqrt{\ }\right)^2 + \frac{\frac{\mathcal{E}_o}{c} - p_{oz}}{\frac{\mathcal{E}_o}{c} + p_{oz}}\right\} \cdot}$$

$$\cancel{\cdot \frac{1}{p_{oz} + qEt + \sqrt{\ }} = -\frac{\mathcal{E}_o}{qE} + \frac{c}{2qE}}$$

$$\left(z + \frac{\mathcal{E}_0}{qE}\right)\cdot\frac{2qE}{c} = \left(\frac{\mathcal{E}_0}{c} + p_{0z}\right)e^{\lambda\tau} + \left(\frac{\mathcal{E}_0}{c} - p_{0z}\right)e^{-\lambda\tau} =$$

$$= p_{0z} + qEt + \sqrt{(p_{0z}+qEt)^2 + \frac{\mathcal{E}_0^2}{c^2} - p_{0z}^2} + \frac{\left(\frac{\mathcal{E}_0}{c} - p_{0z}\right)\left(\frac{\mathcal{E}_0}{c} + p_{0z}\right)}{p_{0z}+qEt+\sqrt{(p_{0z}+qEt)^2+\frac{\mathcal{E}_0^2}{c^2}-p_{0z}^2}} =$$

$$= \frac{(p_{0z}+qEt)^2 + (p_{0z}+qEt)^2 + \frac{\mathcal{E}_0^2}{c^2} - p_{0z}^2 + \frac{\mathcal{E}_0^2}{c^2} - p_{0z}^2 + 2(p_{0z}+qEt)\sqrt{}}{p_{0z}+qEt+\sqrt{(p_{0z}+qEt)^2+\frac{\mathcal{E}_0^2}{c^2}-p_{0z}^2}} =$$

$$= 2\,\frac{(p_{0z}+qEt)^2 + \frac{\mathcal{E}_0^2}{c^2} - p_{0z}^2 + (p_{0z}+qEt)\sqrt{(p_{0z}+qEt)^2+\frac{\mathcal{E}_0^2}{c^2}-p_{0z}^2}}{p_{0z}+qEt+\sqrt{(p_{0z}+qEt)^2+\frac{\mathcal{E}_0^2}{c^2}-p_{0z}^2}} =$$

$$= 2\sqrt{(p_{0z}+qEt)^2+\frac{\mathcal{E}_0^2}{c^2}-p_{0z}^2}\;\frac{\left\{\sqrt{(p_{0z}+qEt)^2+\frac{\mathcal{E}_0^2}{c^2}-p_{0z}^2} + p_{0z}+qEt\right\}}{p_{0z}+qEt+\sqrt{(p_{0z}+qEt)^2+\frac{\mathcal{E}_0^2}{c^2}-p_{0z}^2}}$$

$$= 2\sqrt{(p_{0z}+qEt)^2+\frac{\mathcal{E}_0^2}{c^2}-p_{0z}^2}$$

Finally

$$\Rightarrow z(t) = -\frac{\mathcal{E}_0}{qE} + \frac{c}{qE}\sqrt{(p_{0z}+qEt)^2 + m^2c^2 + p_{0x}^2 + p_{0y}^2} =$$

$$= \frac{c}{qE}\left\{\sqrt{(p_{0z}+qEt)^2 + m^2c^2 + p_{0x}^2 + p_{0y}^2} - \frac{\mathcal{E}_0}{c}\right\}.$$

While

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \frac{1}{m}\begin{pmatrix} p_{0x} \\ p_{0y} \end{pmatrix}\tau = \frac{c}{qE}\begin{pmatrix} p_{0x} \\ p_{0y} \end{pmatrix}\ln\left[\frac{p_{0z}+qEt+\sqrt{(p_{0z}+qEt)^2+m^2c^2+p_{0x}^2+p_{0y}^2}}{\frac{\mathcal{E}_0}{c}+p_{0z}}\right]$$

$$z(t) = \frac{c}{qE}\left\{\sqrt{m^2c^2 + p_{ox}^2 + p_{oy}^2 + p_{oz}^2 + 2p_{oz}\cdot qEt + (qEt)^2} - \frac{\mathcal{E}_0}{c}\right\} =$$

$$= \frac{c}{qE}\left\{\sqrt{\frac{\mathcal{E}_0^2}{c^2} + 2p_{oz}\cdot qEt + (qEt)^2} - \frac{\mathcal{E}_0}{c}\right\}$$

Short-time expansion: $qEt \ll \frac{\mathcal{E}_0}{c}$

$$z(t) \approx \frac{c}{qE}\left\{\frac{\mathcal{E}_0}{c}\sqrt{1 + \frac{2c^2 p_{oz}\, qEt}{\mathcal{E}_0^2} + \left(\frac{c\,qEt}{\mathcal{E}_0}\right)^2} - \frac{\mathcal{E}_0}{c}\right\}$$

$$= \frac{c}{qE}\,\frac{\mathcal{E}_0}{c}\left[\frac{c^2 p_{oz}\, qEt}{\mathcal{E}_0^2} + \frac{1}{2}\left(\frac{c\,qEt}{\mathcal{E}_0}\right)^2 - \frac{1}{2}\left(\frac{c^2 p_{oz}\, qEt}{\mathcal{E}_0^2}\right)^2\right] =$$

$$= \frac{c}{qE}\left[\frac{c\,p_{oz}\, qEt}{\mathcal{E}_0} + \frac{\mathcal{E}_0}{2c}\left(\frac{c\,qEt}{\mathcal{E}_0}\right)^2\left[1 - \left(\frac{c\,p_{oz}}{\mathcal{E}_0}\right)^2\right]\right]$$

$$\frac{\mathcal{E}_0}{c} = \frac{1}{c}\sqrt{m^2c^4 + c^2p_0^2} = \sqrt{m^2c^2 + p_0^2} \approx mc \text{ if initial momentum } p_0 \ll mc$$

$$\Rightarrow z(t) \approx \frac{c}{qE}\left\{\frac{p_{oz}\, qEt}{mc} + \left(\frac{qEt}{mc}\right)^2 \frac{mc}{2}\left[1 - \left(\frac{p_{oz}}{mc}\right)^2\right]\right\}$$

$$= \frac{p_{oz}\, t}{m} + \frac{qEt^2}{2m} \Rightarrow \text{non-relativistic motion with acceleration } \frac{qE}{m}.$$

Long-time expansion: $qEt \gg \frac{\mathcal{E}_0}{c}$

$$\Rightarrow z(t) \approx \frac{c}{qE}\, qEt = ct.$$

$$x(t) \approx \frac{c\,p_{ox}}{qE}\ln\frac{2qEt}{mc}$$

In general, the motion is hyperbolic.